Attention-Based Safe Reinforcement Learning with a Feasibility-Preserving Safety Layer for Size-Agnostic V2G Charging Control

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[TODO: The author chooses the final title; this is the working title.]

Abstract

Charge point operators that offer vehicle-to-grid (V2G) services must keep a station within its transformer limit and deliver the agreed energy to each driver by departure. Reinforcement learning (RL) controllers usually treat both requirements as reward penalties, and their networks are tied to one station size. This paper combines a size-agnostic attention policy with a safety layer, an exact projection onto per-vehicle power bounds and the two-sided transformer limit, computed in $O(N \log N)$ time for N chargers. Each charger is a token, so one set of weights serves any number of chargers.

The lower bound of each electric vehicle (EV) is the least power after which full-power charging still reaches its admitted requirement, and it includes the charging taper. This admitted requirement is the driver's request capped to what is reachable and admissible.

For any sequence of raw actions, during training and deployment, the paper proves two results. Under simulator and grid-headroom conditions, the transformer limit holds at every step. In addition, every EV whose stay contains no infeasible step meets its admitted requirement, where a step is infeasible if the transformer cannot give every EV its lower bound. A least-laxity-first fallback rule handles the remaining steps.

For undeclared departures, a split-conformal bound on the departure time extends the second result with marginal coverage of at least the nominal level minus the rate of EVs that receive less than their bound. The safe attention-based controller (SA-Attn) is trained with proximal policy optimization (PPO) on stations with 10 and 20 chargers in the EV2Gym simulator and tested without retraining on up to 200 chargers. It is compared with learned, constrained-RL, heuristic and model-predictive baselines through the gap of a penalised profit to a perfect-information optimum and through the share of EVs without infeasible steps. [RESULT: one or two sentences with the main findings: penalised-profit gap at $N = 200$, fraction of EVs whose stay contains no infeasible step, cost of the guarantee]

Keywords: Electric vehicle charging, Safe reinforcement learning, Safety layer, Vehicle-to-grid, Zero-shot transfer

1 Introduction

A charge point operator (CPO) buys energy for its chargers at time-varying prices, and vehicle-to-grid (V2G) discharging lets the same operator sell energy from parked electric vehicles (EVs) when prices are high. Two hard requirements limit this freedom. The aggregate power of the station must stay within the transformer limit, and each EV must hold its agreed energy when it leaves. Otherwise the operator overloads equipment or breaks the service promise to the driver.

Reinforcement learning (RL) has become a common approach to this control task, because prices, arrivals and departures are uncertain and hard to model in advance [1, 2]. Most RL controllers for EV charging encode the departure energy requirement as a weighted penalty in the reward [3–6]. Those that model a transformer limit usually penalise overloads in the same way [6, 7]. A penalty trades violations against profit, so a trained policy may still violate either requirement at test time [6,8]. Constrained formulations

bound the expected cumulative cost instead [9], but such a bound holds on average over episodes and not at every step [10].

A second limitation concerns station size, because a flat observation with one action per charger, as in the default interface of the EV2Gym simulator [6], ties a trained network to one number of chargers. Graph and attention policies remove this restriction [7, 11, 12], yet in these size-agnostic controllers the constraints still enter through reward terms, imitation data or heuristic filters.

Safety layers correct each action before execution and can enforce limits by construction [8, 13, 14]. For EV stations, existing layers cover per-step limits such as transformer capacity and charger current [15], or they replace unsafe actions [16, 17]. A per-step layer alone does not protect the departure energy requirement, because that requirement couples the present action with every later step of the stay. Per-EV latest-start bounds address this coupling in laxity-based scheduling [18] and in a recent hierarchical V2G scheduler [19]. Section 2.5 positions the combination of these elements against the closest works.

This paper proposes a safe attention-based controller (SA-Attn) for V2G charging control at a CPO station. An attention policy treats each charger as a token and outputs one raw power action per token with shared weights. A safety layer, an exact projection onto the feasible set, maps the raw actions to commanded powers that respect per-EV bounds and the transformer limit. The projection onto a box intersected with a two-sided sum constraint is classical [20, 21], and the rate rule of smoothed least-laxity-first (sLLF) scheduling has a closely related form [18]. Our feasibility argument extends the laxity argument of Chen et al. to the charging taper, V2G discharging and a two-sided coupling.

The paper makes four contributions.

1. We derive a safety layer whose per-EV lower bound includes the charging taper, and Theorem 1 states its result in two layers. The transformer limit holds at every step for every sequence of raw actions under simulator and grid-headroom conditions, and the admitted departure requirement is met by every EV whose stay contains no infeasible step.
2. We use a charger-token attention policy as the size-agnostic vehicle of the layer and train it with the layer in the loop. Variants without the departure bound, with the layer at deployment only and with a Deep Sets policy show what each part contributes.
3. We evaluate the controller in EV2Gym with training on 10 and 20 chargers and zero-shot tests on up to 200 chargers and on a tighter transformer rating. The main metric is a penalised-profit gap to a perfect-information optimum that bounds every compared method. A primary endpoint is the share of EVs without infeasible steps, and a hard-constrained optimum supports the cost of the guarantee.
4. As a secondary contribution, a split-conformal lower bound on the departure step replaces the known-departure assumption, and Proposition 2 turns the departure result into a coverage statement.

None of these ingredients is new in isolation, so the claim of the paper rests on their combination, on the two-layer result and on its measured cost.

Section 2 reviews related work, Section 3 states the model as a partially observable Markov decision process (POMDP), and Section 4 presents the method. Sections 5 to 8 cover the setup, results, discussion and conclusion, and the supplementary material holds all proofs and further details.

2 Related Work

2.1 Reinforcement learning for EV charging and V2G

Deep RL for EV charging has moved from single-vehicle agents [22] to stations with photovoltaics, storage and V2G [5, 17], to coordinated stations [23] and to fleets under distribution-grid limits [24, 25]. Two reviews list safety, dependence on simulation and the lack of a standard benchmark among the open problems [1, 2]. EV2Gym models a CPO whose chargers sit under transformers with inflexible load and solar generation [6]. Its V2G reward penalises transformer overload and low user satisfaction, and

in its V2G benchmark penalty-based agents earned profit but left many users short of their requested energy. The Adaptive Charging Network (ACN) solves a convex program in receding horizon with hard infrastructure limits and best-effort energy delivery [26]. An economic model predictive control (MPC) inside EV2Gym enforces hard departure and transformer constraints [27].

2.2 Constraint handling and safe reinforcement learning

Weighted violation terms in the reward do not prevent violations at test time [6, 8]. The constrained Markov decision process (CMDP) bounds an expected cumulative cost instead [9], solved by Lagrangian [28] or trust-region [29] updates. Reviews note that such bounds hold for an expected cumulative cost rather than at each step [10] and classify Lagrangian methods as soft guarantees [30]. CMDP formulations have been applied to EV charging with voltage limits [31], transformer thermal stress [32, 33] and V2G aggregators [34]. A Lagrangian multi-agent method also encodes the recent history of each station with a transformer [35].

A second family corrects each action before execution, through shields [36], learned linear safety layers [13] or quadratic programs with a fallback controller [37]. EV examples include rule-based shields [16], action-replacement layers [17], grid-code shielding [38] and power-flow shields [39]. Such a filter is only as safe as its internal model, and the authors of the HetGPS preprint make no pointwise feasibility claim for its learned risk model with a linearised power-flow projection [12]. Members of the EV2Gym team have also combined a graph-enhanced twin delayed deep deterministic policy gradient (TD3) agent [40] with a safety layer [41]. According to the chapter abstract, the layer is intended to preserve charging requirements and enforce power limits.

A third family makes the policy output feasible by construction. Differentiable convex optimisation layers pass gradients through an exact solve [42, 43], and Chen et al. used such a projection inside energy-system policies [14]. Gauge maps give feasible points directly [44], and Tabas and Zhang used one to prove that a learned frequency controller stays in a controlled-invariant safe set [8]. For EV stations, a preprint by Wan et al. proves per-step feasibility of a quadratic-program projection that covers transformer and charger-current limits but not departure energy [15]. A preprint by Yan et al. derives per-EV bounds from remaining energy and time, and its formal result concerns first-order power-balance feasibility [19]. Projection also alters learning, because naive projection biases policy gradients [45] and an active projection has zero gradients [46]. For stochastic policies, a recent preprint shows that projection inside the environment is equivalent to projection as a policy layer [47].

2.3 Size-agnostic architectures

Deep Sets [48], self-attention [49] and the Set Transformer [50] process sets of any size with shared weights, and UPDeT applied this idea to multi-agent tasks of different sizes [51]. Size transfer is not automatic, since neural routing solvers trained on small graphs often fail on larger ones [52]. Early size-independent states in EV charging came from aggregation, and Sadeghianpourhamami et al. also reported the gap to a perfect-knowledge optimum under scaled arrival rates [3]. Aggregated representations, however, hide per-vehicle detail or move it into disaggregation rules [53, 54].

The preprint version of the heterogeneous graph policy (EV-GNN) coordinates up to 1000 charge points, with a separate model per scale and no constraint enforced by construction [11]. In contrast, the graph-enhanced decision transformer (GNN-DT), trained on 25 chargers, was tested without retraining on 5 to 100 chargers, and its performance degraded as size grew [7]. A HetGPS policy trained with 1,236 vehicles transfers without retraining to feeders with 2,239 and 3,218 vehicles [12]. Similarly, a graph actor for voltage control has been evaluated in simulation on a larger feeder than in training [55].

In these size-agnostic controllers, constraints enter through reward terms, imitation data or heuristic filters.

2.4 Laxity-based scheduling and departure uncertainty

Chen et al. proved that a schedule is feasible exactly when every laxity, the slack between the time left and the time still needed to finish charging, stays non-negative [18]. Their sLLF rule clips each rate

Table 1: Positioning against the closest works. "Penalty" means a reward term; "hard" means enforced by construction.

Work	Size-agnostic policy	Departure energy	Transformer limit	Guarantee	Zero-shot size test	Gap to optimum
EV2Gym agents [6]	no (flat)	penalty	penalty	none	no	optimum as baseline
EV-GNN, preprint version [11]	yes (graph)	not by construction	not by construction	none	separate model per scale	optimisation baseline per scale
GNN-DT [7]	yes (graph)	penalty	penalty	none	25 to 5–100 chargers	near-oracle reward (journal abstract)
HetGPS, preprint [12]	yes (graph attention)	reward and heuristic	not constrained	empirical (voltage filter)	1,236 to 3,218 vehicles	no
Yan et al., preprint [19]	no	hard (per-EV rule)	not modelled	first-order power balance	no	gap to MPC
Wan et al., preprint [15]	no	not modelled	hard (quadratic-program projection)	per-step feasibility	no	no
sLLF [18]	rule, any size	laxity rule	hard (common level)	feasibility theorems	not learned	no
Sadeghian-pourhamami et al. [3]	yes (aggregate state)	penalty	not modelled	none	scaled arrivals	yes
Orfanoudakis et al. [41] (abstract)	graph policy	safety layer (stated aim)	safety layer (stated aim)	not stated in abstract	not stated in abstract	not stated in abstract
This work	yes (charger tokens)	per-EV bound, conditional (Thm. 1b)	hard (common shift)	transformer: every step (A3, A6); departure: per EV, conditional	10/20 to 200 chargers	soft- and hard-constrained optima

around a common level under the station power limit, and their bounds indicate that online feasibility generally requires extra power. A preprint shows that hard deadlines and a transformer limit can become jointly infeasible under congestion [56], and least-laxity-first (LLF) priorities perform well under cable and transformer limits [57, 58]. These methods assume that users declare departure time and energy at arrival. Plug-in durations are hard to predict [59], users can gain by under-reporting deadlines [60], and departure forecasts improve prioritisation [61]. A recent preprint trains a departure forecaster end-to-end with the charging policy but keeps unmet energy as a soft penalty and gives no coverage statement [62]. Conformal prediction gives distribution-free coverage under exchangeability [63, 64] and has tightened constraints in safe planning [65].

2.5 Positioning

As seen in Table 1, each ingredient of our design already exists in some form. Size-agnostic policies with zero-shot transfer exist for EV charging [7,12], provable action-correction layers exist in safe RL [8,15], and a closely related clip-around-a-common-level rule appears in sLLF [18]. Per-EV latest-start bounds appear in the scheduler of Yan et al. [19], and train-small, test-large evaluation with a gap to an optimum dates back at least to [3]. Three closely related recent works were available to us only as abstracts [23, 34, 41], and we describe them only at that level. To our knowledge, no prior work joins a size-agnostic learned policy, a per-step transformer guarantee and a multi-step departure argument in one controller.

3 Problem Formulation

3.1 Station, EVs and energy dynamics

A CPO runs a station with N chargers $\mathcal{N} = \{1, \dots, N\}$ behind one transformer, and each charger hosts at most one EV, whose quantities carry the index i of its charger. Time is divided into steps $t = 0, \dots, T-1$ of length Δt [h]. At every step the station observes the transformer rating $P_{\mathrm{tr}}(t)$, the inflexible load $P^{\mathrm{inf}}(t) \geq 0$ and the photovoltaic (PV) generation $P^{\mathrm{pv}}(t) \geq 0$ in kW. Energy is bought at the price $c^{\mathrm{ch}}(t)$ and sold at the price $c^{\mathrm{dis}}(t)$ [€/kWh], and both prices may be negative.

EV i arrives at step t_i^{arr} and is controlled until its departure step d_i, the last controlled step, so its departure energy is $E_i(d_i + 1)$, with $E_i(t)$ the battery energy at the start of step t. At step t the EV has $\tau_i(t) = d_i - t + 1$ remaining actions, and $\mathcal{N}(t)$ is the set of the $n(t)$ occupied chargers. Let $p_i(t)$ [kW] be the power commanded at charger i, positive for charging and negative for V2G discharging, and let $\tilde{p}_i(t)$ be the power reported by the simulator. With $p^+ = \max(p, 0)$ and $p^- = \max(-p, 0)$, the battery energy evolves as

$$E_i(t+1) = E_i(t) + \Delta t \, g_i\big(p_i(t)\big), \qquad g_i(p) = \eta_i^{\mathrm{c}} p^+ - p^- / \eta_i^{\mathrm{d}}, \tag{1}$$

with charge and discharge efficiencies $\eta_i^{\mathrm{c}}, \eta_i^{\mathrm{d}} \in (0, 1]$. Per-step physical limits restrict the power to

$$p_i^{\min}(E) \leq p_i(t) \leq \bar{P}_i(E), \qquad p_i^{\min}(E) = -\min\Big\{P_i^{\mathrm{dis}},\ \eta_i^{\mathrm{d}} \frac{(E - E_i^{\min})^+}{\Delta t}\Big\}, \qquad E = E_i(t), \tag{2}$$

where the maximum charging power $\bar{P}_i(E) \geq 0$ includes the charging taper and the capacity limit, $\bar{P}_i^{\mathrm{nom}} = \sup_E \bar{P}_i(E)$ and P_i^{dis} is the maximum discharging power. The theory holds for the general map g_i, but EV2Gym reports, bills and meters the battery-side energy change as the charger power [6]. Our experiments therefore use $\eta_i^{\mathrm{c}} = \eta_i^{\mathrm{d}} = 1$, a convention that removes the round-trip loss of V2G arbitrage (Section 7.3).

3.2 Departure requirement and transformer limit

Each EV requests a desired departure energy E_i^{des}, and a station-level request pipeline turns it into the admitted requirement E_i^{req}. The pipeline runs once per test replay before any controller, and every compared method and both optima receive the same admitted requirement. At arrival it caps the request to the energy that full-power charging can reach during the stay, about 10^{-6} kWh below that value. Admission control then solves a small feasibility linear program (LP) and lowers the new request if the connected EVs could otherwise not all be served under the forecast transformer limits. Requests already admitted are never reduced. Hence

$$E_i^{\mathrm{req}} \leq \min\big\{E_i^{\mathrm{des}},\ \text{the energy reachable at full power from } E_i^{\mathrm{arr}} \text{ during the stay}\big\}, \tag{3}$$

and Section S3 gives the details. The departure requirement is

$$E_i(d_i + 1) \ \geq \ E_i^{\mathrm{req}}. \tag{4}$$

Without the cap, some arrivals could not be served by any controller, and a guarantee about them would be void. For every method, shortfall is measured against the shared admitted requirement, and the energy removed by the pipeline is reported separately.

The net transformer power $P^{\mathrm{inf}}(t) - P^{\mathrm{pv}}(t) + \sum_i p_i(t)$ must stay within $[-P_{\mathrm{tr}}(t), P_{\mathrm{tr}}(t)]$, so the EV aggregate has the transformer limit

$$-C_{\mathrm{rev}}(t) \leq \sum_{i \in \mathcal{N}} p_i(t) \leq C(t), \qquad C(t) = P_{\mathrm{tr}}(t) - P^{\mathrm{inf}}(t) + P^{\mathrm{pv}}(t), \qquad C_{\mathrm{rev}}(t) = P_{\mathrm{tr}}(t) + P^{\mathrm{inf}}(t) - P^{\mathrm{pv}}(t). \tag{5}$$

The forward limit $C(t)$ bounds charging and the reverse limit $C_{\mathrm{rev}}(t)$ bounds V2G export. For the current step the safety layer uses the measured load and PV, whereas the policy observes only noisy forecasts of future values.

3.3 Online control problem and optima

At each step the CPO chooses $p(t) \in \mathbb{R}^N$ using only the information available at step t. The profit of a controller π in an episode, in €, is

$$J^{\pi} = \sum_{t=0}^{T-1} \Delta t \sum_{i \in \mathcal{N}(t)} \big(c^{\mathrm{dis}}(t)\, \tilde{p}_i^{-}(t) - c^{\mathrm{ch}}(t)\, \tilde{p}_i^{+}(t) \big), \tag{6}$$

and the controller should maximise its expectation subject to Equations 1–2, Equation 4 for every EV and Equation 5 at every step. With the weights $w_E = w_P = 10\,€/\mathrm{kWh}$ for every method, the penalised profit is

$$J^{\mathrm{pen}}(\pi) = J^{\pi} - w_E \sum_i \Delta E_i^{\mathrm{sh}} - w_P\, \Delta t \sum_t \Big[\Big(\sum_i \tilde{p}_i(t) - C(t) \Big)^{+} + \Big(-C_{\mathrm{rev}}(t) - \sum_i \tilde{p}_i(t) \Big)^{+} \Big], \tag{7}$$

where $\Delta E_i^{\mathrm{sh}} = (E_i^{\mathrm{req}} - E_i(d_i + 1))^{+}$ is the departure shortfall.

Two perfect-information LPs, solved with HiGHS [66], know the whole episode and serve as references. Both model EV2Gym's taper with outer linear rows, exact except on a one-step transition band and a 1% margin (Lemma S8), and both allow for EV2Gym's unbilled 0.01 kWh rounding gain. The soft-penalty optimum $J^{\mathrm{pen,LP}}$ maximises Equation 7 with slack variables for shortfall and overload. For every simulated trajectory, with shortfall measured against the shared admitted requirement, $J^{\mathrm{pen}}(\pi) \le J^{\mathrm{pen,LP}}$ (Proposition S3), including trajectories with shortfall or overload. The main metric is therefore well defined for every compared method,

$$\mathrm{gap}^{\mathrm{pen}}(\pi) = \frac{J^{\mathrm{pen,LP}} - J^{\mathrm{pen}}(\pi)}{|J^{\mathrm{pen,LP}}|} \ge 0, \tag{8}$$

where the absolute value is needed because profits are usually negative. Episodes with $|J^{\mathrm{pen,LP}}| < 1\,€$ or a non-optimal solver status get no gap, and their number is reported. The hard optimum $J^{\mathrm{LP,ceil}}$ enforces the departure requirements and the transformer limit as constraints, after a first stage that minimises the total shortfall. It bounds the profit of every controller that respects the transformer limit and leaves no more shortfall (Proposition S2), and it serves the analysis of the cost of the guarantee.

3.4 Partially observable Markov decision process

The online problem is a POMDP, because future arrivals, prices beyond a short window and future loads are not observed. We act on the current observation with a feed-forward actor and critic. The policy outputs one raw action $a_i(t) \in \mathbb{R}$ per charger, scaled to kW as $x_i(t) = P^{\mathrm{ref}} a_i(t)$ with $P^{\mathrm{ref}} = 11$ kW, and the safety layer of Section 4.3 maps it to the commanded power without any squashing function. Its reward is the realised profit of the step, with no penalty for the departure requirement or the transformer limit, because the safety layer enforces both. For training, the reward is divided by $N r^{\mathrm{ref}}$ with $r^{\mathrm{ref}} = c^{\mathrm{ref}} P^{\mathrm{ref}} \Delta t$ and $c^{\mathrm{ref}} = 0.25\,€/\mathrm{kWh}$, so that returns do not grow with N, and all reported results use the undivided profit. Arrivals, prices, loads and PV evolve independently of the actions, every EV departs before the episode ends, and the discount factor $\gamma = 0.99$ gives an effective horizon of about one day at $\Delta t = 15$ min.

4 Proposed Method

As seen in Figure 1, SA-Attn has three parts that act in sequence at every step. An attention policy maps a set of charger tokens to one raw action per charger, and a safety layer turns the raw actions into commanded powers that respect per-EV bounds and the transformer limit. During training, a policy-gradient method updates the policy from the realised profit with the safety layer treated as part of the environment, and the same layer acts unchanged during deployment.

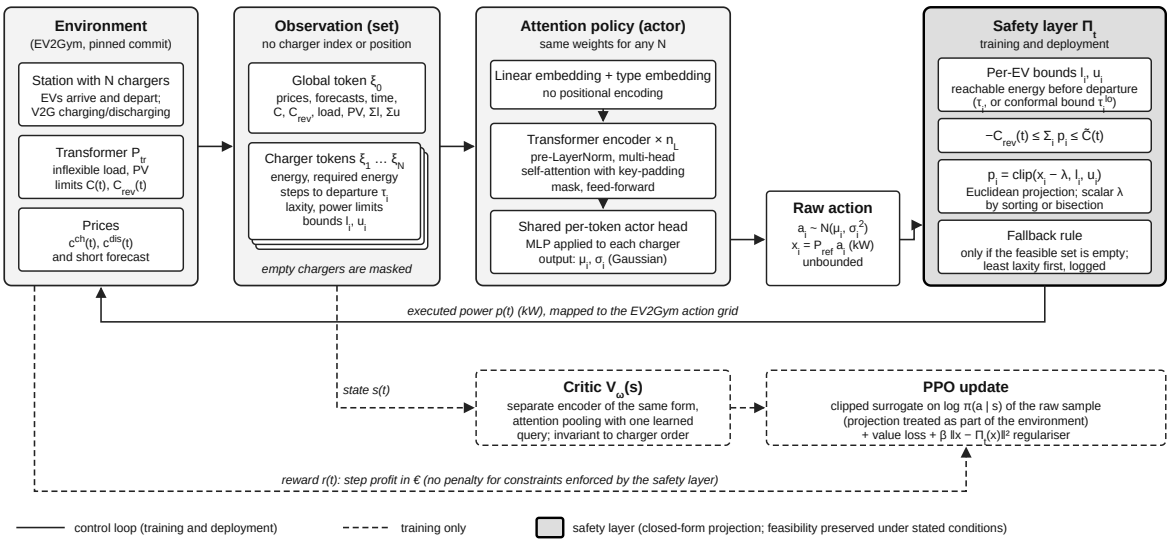


Figure 1: Architecture and control loop of SA-Attn. Solid arrows form the control loop used in training and deployment; dashed arrows are used in training only. The safety layer (shaded) is an exact projection that keeps the transformer limit and keeps each EV's requirement reachable while the EV receives its lower bound (Theorem 1).

4.1 Token representation and attention policy

The observation is a set of N charger tokens $\xi_i(t) \in \mathbb{R}^{12}$ and one global token $\xi_0(t) \in \mathbb{R}^{38}$, and no feature encodes the index or the position of a charger. A charger token holds occupancy, energy and capacity data, the remaining required energy and steps, the laxity of Section 4.2, the power limits and the bounds l_i and u_i. The global token holds prices for the next $H_f = 8$ steps, time features, the current limits, load and PV forecasts and four station ratios. All features are normalised so that they do not grow with N, and Table S3 lists them.

Each token is embedded linearly with one of two learned type embeddings, $e_i = W_c\xi_i + b_c + v_c$ for chargers and $e_0 = W_g\xi_0 + b_g + v_g$ for the global token. The sequence $[e_0; e_1; \dots; e_N]$ passes through $n_L = 3$ pre-LayerNorm transformer encoder layers [49, 67, 68] of width $d_m = 128$ with $n_H = 4$ heads and no positional encoding. A key-padding mask removes empty chargers as attention keys, and the global key is never masked. Then a shared actor head maps each charger output h_i to the mean $\mu_{\theta,i}$ and standard deviation $\sigma_{\theta,i}$ of a Gaussian raw action, where θ denotes the actor parameters. The policy is $\pi_\theta(a \mid s) = \prod_{i\in\mathcal{N}(t)} \mathrm{Normal}\big(a_i; \mu_{\theta,i}(s), \sigma^2_{\theta,i}(s)\big)$. A separate critic encoder with parameters ω pools all unmasked token outputs with one learned attention query [50] into the value $V_\omega(s)$.

The construction is permutation-equivariant, masked tokens do not influence the outputs of unmasked tokens, and the number of parameters does not depend on N (Lemma S1). Whatever the number of chargers, the actor has 610,370 parameters and the critic 676,481. Weights trained at $N \in \{10, 20\}$ can therefore run at $N = 200$, but performance there is not guaranteed, because attention over 200 tokens averages over more keys than seen in training.

4.2 Per-EV bounds from reachable energy

For each connected EV, the safety layer needs the least power that keeps its departure requirement reachable, and we drop the index i where no confusion arises. Full-power charging for one step maps the energy E to $\varphi(E) = E + \Delta t\, \eta^{\mathrm{c}}\, \bar{P}(E)$. Assumption A1 of Section 4.4 asks that φ is continuous and non-decreasing, so that starting a step with more energy never leaves less energy after full-power charging. The backward-reachable thresholds, the least energies from which the requirement is still reachable at full power, are

$$B(0) = E^{\mathrm{req}}, \qquad B(k) = \min\big\{E \in [E^{\min}, E^{\mathrm{cap}}] :\ \varphi(E) \geq B(k-1)\big\}, \quad k \geq 1. \tag{9}$$

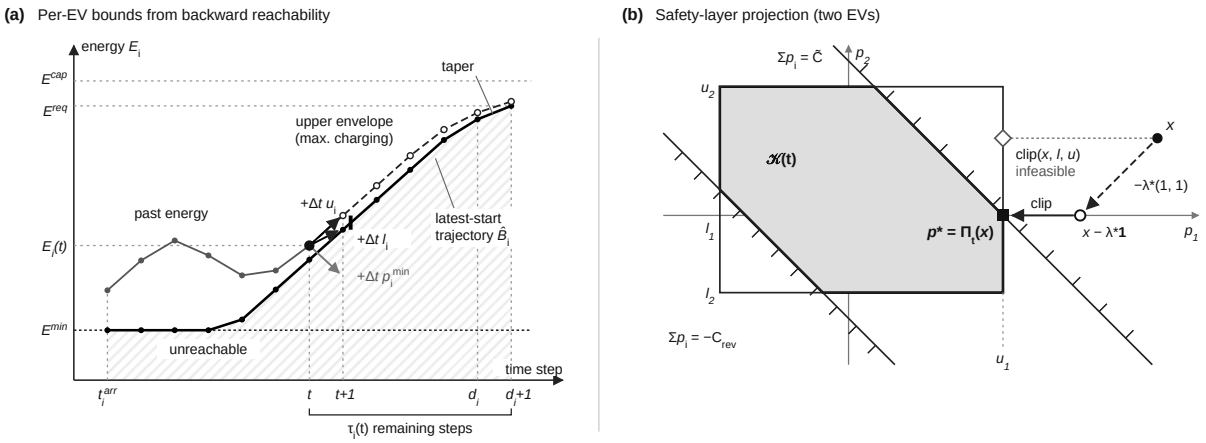


Figure 2: Concepts of the safety layer. (a) Energy of one EV during its stay, with the latest-start trajectory $\hat{B}_i$ and the per-step options $\Delta t\, u_i$, $\Delta t\, l_i$ and $\Delta t\, p_i^{\min}$. (b) Projection for two EVs onto the box $[l, u]$ cut by the slab $-C_{\mathrm{rev}} \le p_1 + p_2 \le \tilde{C}$.

Under A1, an EV is individually feasible at step t, in the sense that some admissible control meets Equation 4, exactly when $E(t) \ge B(\tau(t))$ (Lemma S2). Without A1 this characterisation fails, because charging less now could leave more reachable energy later. For EV2Gym's exponential taper, Lemma S6 proves A1 for every step length.

For a connected EV with energy $E = E_i(t)$ and $\tau = \tau_i(t)$ remaining actions, the bounds of the safety layer are

$$u_i(t) = \bar{P}(E), \qquad l_i(t) = \begin{cases} \min\Big\{u_i(t),\ \max\Big\{p^{\min}(E),\ g^{-1}\Big(\dfrac{\hat{B}(\tau-1) - E}{\Delta t}\Big)\Big\}\Big\}, & E \ge \hat{B}(\tau), \\ \bar{P}(E), & E < \hat{B}(\tau), \end{cases} \tag{10}$$

where $\hat{B}$ is a stored threshold table and the outer minimum is a numerical guard. In the first case, l_i is the smallest admissible power after which $\tau - 1$ full-power steps still reach E^{req}, and an individually infeasible EV is charged at maximum power. A negative lower bound allows V2G discharging while slack remains, and as seen in Figure 2(a), the bound keeps the energy on or above the latest-start trajectory.

The table $\hat{B}_i(k)$, $k = 0, \dots, \tau_i(t_i^{\mathrm{arr}})$, is computed once at arrival by bisection on φ_i, so the bounds cost $O(1)$ per EV and step. Each stored table is certified in the sense that

$$\hat{B}(0) \ge E^{\mathrm{req}}, \qquad \varphi\big(\hat{B}(k)\big) \ge \hat{B}(k-1), \qquad \hat{B}(k) \le \max\{\hat{B}(k-1), E^{\min}\}, \qquad \hat{B}(k) \in [E^{\min}, E^{\mathrm{cap}}], \tag{11}$$

and the proofs use only these properties. The laxity $\chi_i(t) = \big(E_i(t) - \hat{B}_i(\tau_i(t))\big) / \big(\eta_i^{\mathrm{c}} \bar{P}_i^{\mathrm{nom}} \Delta t\big)$, measured in steps, is non-negative exactly when the EV is individually feasible.

4.3 Projection

Let $x \in \mathbb{R}^N$ be the raw action in kW, with $x_i = l_i = u_i = 0$ for empty chargers. The safety layer uses the tightened forward limit $\tilde{C}(t) = C(t) - n(t)\varepsilon_q$, where the margin $\varepsilon_q = 0.008$ kW per charger absorbs the actuation rounding of the simulator (Section S2.2). Its feasible set is $\mathcal{K}(t) = \{p \in \mathbb{R}^N :\ l \le p \le u,\ -C_{\mathrm{rev}}(t) \le \mathbf{1}^\top p \le \tilde{C}(t)\}$, and the safety layer is the Euclidean projection

$$\Pi_t(x) = \arg\min_{p \in \mathcal{K}(t)} \tfrac{1}{2}\|p - x\|_2^2. \tag{12}$$

Proposition 1 (Projection)**.** *(i) The set $\mathcal{K}(t)$ is non-empty if and only if $l \le u$, $\sum_i l_i \le \tilde{C}$, $\sum_i u_i \ge -C_{\mathrm{rev}}$ and $-C_{\mathrm{rev}} \le \tilde{C}$.*

(ii) If $\mathcal{K}(t) \ne \emptyset$, the unique minimiser of Equation 12 is $p_i^* = \mathrm{clip}(x_i - \lambda^*, l_i, u_i)$. The common shift λ^* is zero if the clipped point already satisfies the aggregate limit, and otherwise it is the root of $\lambda \mapsto \sum_i \mathrm{clip}(x_i - \lambda, l_i, u_i)$ at the violated limit.

The proof verifies the Karush–Kuhn–Tucker (KKT) conditions, and the supplementary material also gives the Jacobian (Proposition S1). As seen in Figure 2(b), the projection shifts all free coordinates by the same amount and clips them until the aggregate limit holds. One sort of the $2N$ breakpoints $\{x_i - u_i\} \cup \{x_i - l_i\}$ and one pass over them give λ^*, so the layer costs $O(N \log N)$ per step. A post-check with a bisection fallback keeps the aggregate limit under floating-point rounding, and Algorithm S1 lists one complete step.

The algebra is not new: Equation 12 is the continuous quadratic knapsack problem [20, 21], and the sLLF rate rule uses a closely related form, a common level scaled per EV by its peak rate [18]. Our use differs in its input, a learned per-charger action of a size-agnostic policy, and in the lower bound l_i from backward reachability with taper and V2G sign range. Moreover, the aggregate constraint is two-sided, and the layer acts during training.

4.4 Feasibility preservation

Intuitively, the safety layer never lets an EV fall below the least energy from which its requirement is still reachable at full power, and the common shift keeps the station inside the transformer band. Only the first property can fail, namely when the transformer cannot give every EV its lower bound at once. We call a step feasible when $\mathcal{K}(t)$ is non-empty, and on an infeasible step a fallback rule replaces the projection. The conditions of the result are collected below, and only the first five are assumptions. Technically, the induction behind it extends the laxity argument of Chen et al. [18] to the taper, V2G discharging, a two-sided coupling and arbitrary raw actions.

Conditions of Theorem 1

Label	Condition	Status in this study
A1	φ continuous and non-decreasing	proved for EV2Gym's taper (Lemma S6)
A2	departure step d_i and requirement E_i^{req} known at arrival	relaxed by the conformal bound (Section 4.6)
A3	one-step model: $E_i(t+1) \geq E_i(t) + \Delta t\, g_i(p_i)$ and $p_i \leq \tilde{p}_i \leq p_i + \varepsilon_q$ for commands within Equation 2	proved for EV2Gym's actuation (Lemma S7); may fail on hardware
A4	feasible arrival: $E_i^{\mathrm{arr}} \geq \hat{B}_i(\tau_i(t_i^{\mathrm{arr}}))$	enforced by the request pipeline
A6	grid headroom: $C(t) \geq N\varepsilon_q$ and $C_{\mathrm{rev}}(t) \geq 0$	exogenous; checked through the fallback counts
A5	feasible step: $\sum_i l_i \leq \tilde{C}$, $\sum_i u_i \geq -C_{\mathrm{rev}}$, $\tilde{C} \geq -C_{\mathrm{rev}}$	a closed-loop event, not assumed by (a) and (b); measured

A6 uses the number of chargers N instead of $n(t)$, so that it does not depend on the policy, and it fails in our scenarios with a probability below 2×10^{-9} per episode (Section S2.3).

Theorem 1 (Feasibility preservation)**.** *Let the commanded powers be $p(t) = \Pi_t(x(t))$ on feasible steps and the output of the fallback rule otherwise, for an arbitrary sequence of finite raw actions, during training or deployment.*

(a) Under A3 and A6, the realised powers satisfy the transformer limit $-C_{\mathrm{rev}}(t) \leq \sum_i \tilde{p}_i(t) \leq C(t)$, and the commanded powers satisfy the physical limits of Equation 2, at every step.

(b) Under A1–A4, every EV i with $p_i(t) \geq l_i(t)$ throughout its stay meets $E_i(d_i{+}1) \geq E_i^{\mathrm{req}}$, in particular every EV whose stay contains no infeasible step.

(c) If, in addition, every step is feasible, every EV meets its departure requirement.

Proof sketch. On feasible steps, Proposition 1(i) gives $p(t) \in \mathcal{K}(t)$, and A3 transfers the tightened band to the realised powers. Under A6, the fallback rule can only meet the case of excess required power described below, and its output keeps the commanded sum at $\tilde{C}(t)$. For (b), a one-step lemma (Lemma S3) carries the invariant $E_i(t) \geq \hat{B}_i(\tau_i(t))$ from arrival, where A4 gives it, through every step with $p_i \geq l_i$, and at the last step it gives $E_i(d_i + 1) \geq \hat{B}_i(0) \geq E_i^{\mathrm{req}}$. Part (c) follows from (b), and Section S1.5 gives the full proof. □

The transformer part depends neither on the policy nor on A1 or A5, whereas the departure part is a per-EV statement. A5 can fail even when a joint plan exists. Two EVs that each need 10 kWh within two 1-h steps under a 10 kW limit can share the capacity equally. A policy that idles in the first step, however, leaves $\sum_i l_i = 20$ kW in the second step. We therefore measure how often EVs meet infeasible steps.

4.5 Fallback rule

On an infeasible step, the fallback rule replaces the projection with a deterministic least-laxity-first allocation. In the relevant case, the required power $\sum_i l_i$ exceeds $\tilde{C}$. Every EV then starts at its physical lower limit $p_i^{\min}$, and the remaining budget up to $\tilde{C}$ goes to the EVs in increasing order of laxity, each up to l_i. This output keeps the transformer limit, and every feasible EV that still receives l_i stays feasible (Lemma S4). Three further cases cover contradictory limits, excess PV export and a negative budget, and A6 excludes all three (Section S1.6). For the EVs that receive less than l_i, the total departure shortfall of an episode is bounded a priori by the accumulated deficit of these steps (Lemma S5). Admission control in the request pipeline makes infeasible steps rarer, but it relies on forecasts and does not enter Theorem 1.

4.6 Conformal lower bound on the departure step

To relax A2 for the departure step, we use split conformal prediction (CP) [63, 64, 69, 70] in a one-sided, quantile-regression form [71]. A gradient-boosting regressor $\hat{q}$ estimates a lower quantile of d_i from the arrival step, the weekday, the arrival energy, the requested energy and the EV model. Calibration draws one session uniformly from each of n_{cal} independently seeded episodes and uses the scores $e_j = \hat{q}(\mathbf{w}_j) - d_j$. With $\hat{Q}_{1-\alpha}$ the $\lceil (n_{\mathrm{cal}}+1)(1-\alpha) \rceil$-th smallest score, the bound $\hat{d}_i^{\mathrm{lo}} = \max\{t_i^{\mathrm{arr}}, \lceil \hat{q}(\mathbf{w}_i) - \hat{Q}_{1-\alpha} \rceil\}$ replaces d_i in the bounds, the request pipeline and the features, and after $\hat{d}_i^{\mathrm{lo}}$ the layer uses one remaining step.

Proposition 2 (Conformal departure guarantee)**.** *Let the calibration sessions and the test session be exchangeable. Then* $\mathbb{P}(d_i \geq \hat{d}_i^{\mathrm{lo}}) \geq 1-\alpha$*, and under A1, A3 and A4 with* $\hat{d}_i^{\mathrm{lo}}$ *in place of* d_i*,* $\mathbb{P}\big(E_i(d_i+1) \geq E_i^{\mathrm{req}}\big) \geq 1 - \alpha - \mathbb{P}\big(\exists\, t \leq d_i :\ p_i(t) < l_i(t)\big)$*.*

Here E_i^{req} is the requirement re-certified with $\hat{d}_i^{\mathrm{lo}}$, and the probability is over the calibration data and a uniformly chosen EV of a test episode, so the guarantee is marginal. Calibration needs data from each test configuration, so the conformal variant is not zero-shot.

4.7 Training algorithm

The policy is trained with proximal policy optimization (PPO) [72] and generalised advantage estimation [73], with the safety layer, the fallback rule and the actuation treated as part of the environment. PPO therefore uses the likelihood ratio of the raw Gaussian sample. The vanilla policy-gradient estimator, equal to the PPO surrogate gradient at the old parameters, is then unbiased for the return of the projected actions. For stochastic policies, projection inside the environment and inside the policy are equivalent in this sense [47], and the choice avoids the zero gradients of an active projection [46]. Tokens with $u_i - l_i \leq 10^{-9}$ kW cannot influence any commanded power and are dropped from the log-probability and the entropy. Fallback steps are masked out of the actor loss but kept for the critic. Many raw actions map to the same projected point, so on feasible steps a regulariser with weight $\beta = 0.01$ keeps the policy mean close to $\mathcal{K}(t)$,

$$\mathcal{L}_{\mathrm{reg}}(\theta) = \beta\, \mathbb{E}_s\Big[\frac{1}{n(t)} \sum_{i \in \mathcal{N}(t)} \Big(\frac{\bar{x}_i - \Pi_t(\bar{x})_i}{P^{\mathrm{ref}}}\Big)^2\Big], \qquad \bar{x} = P^{\mathrm{ref}} \mu_\theta(s). \tag{13}$$

By the Moreau identity, the gradient of $\frac{1}{2}\mathrm{dist}^2(x, \mathcal{K})$ is $x - \Pi_t(x)$, so $\Pi_t(\bar{x})$ is evaluated under stop-gradient [14, 47]. Each rollout runs eight environments with $N = 10$ and eight with $N = 20$, and Algorithm S2 gives the full training loop.

5 Experimental Setup

5.1 Simulator and scenarios

All experiments use EV2Gym [6] at commit `55ef368c` without changes to its source. A wrapper computes the observations, applies the shared admitted requirements, the safety layer, the fallback rule and the actuation, and logs all metrics without reading future arrivals. Each station has N chargers behind one transformer, and EVs are drawn from the eight V2G-capable EV models of the simulator with one constant efficiency per model (0.90, and 0.93 for one model). The desired energy is the full battery, prices are the Dutch day-ahead prices shipped with EV2Gym, and the discharge price equals the charging price. Table S2 lists all scenario parameters.

The grid quantities and the arrival rate scale linearly with N. In particular, the transformer rating is $P_{\mathrm{tr}} = \rho_{\mathrm{tr}} N$, with a tight level of $\rho_{\mathrm{tr}} = 4$ kW per charger as in EV2Gym's 25-charger V2G configuration and a moderate level of 7 kW. Inflexible load and PV are scaled with multipliers of mean $0.4 P_{\mathrm{tr}}$, and forecasts in the observation carry Gaussian noise with a standard deviation of 10%. The workplace, public and private arrival profiles of EV2Gym start at hours that let overnight sessions fit into the 28-h horizon of $T = 112$ steps of 15 min. Their mean charger occupancy is about 0.21 for every profile.

Training mixes $N \in \{10, 20\}$, the three profiles and the two ratings within every batch. The test grid covers $N \in \{5, 10, 20, 50, 100, 200\}$, both ratings and all profiles, plus a stress level of $\rho_{\mathrm{tr}} = 3$ kW at $N \in \{20, 200\}$. Zero-shot therefore refers to the station size and to the stress level, whereas the profiles are seen in training. Each configuration has 100 fixed test episodes for $N \leq 50$ and 50 for $N \geq 100$, shared by all methods, and EV2Gym's original 25-charger configuration adds 100 episodes, so each policy runs 3,550 test episodes. Per-charger quantities do not change with N, so aggregate fluctuations shrink with growing N, and we report per N how often the aggregate limits bind. Sensitivity studies at $N = 20$ and $\rho_{\mathrm{tr}} = 4$ kW cover a desired state of charge of 0.8, a 5-min step, a throughput cost of 0.05 €/kWh and a discharge price factor of 0.9. Only the 5-min study retrains.

5.2 Compared methods

Every method and every evaluation pass receives the same admitted requirements, and shortfall is always measured against them. Methods without the layer execute only the physical clip $\mathrm{clip}(x_i, p_i^{\min}, u_i)$ without l_i or the transformer coupling, and every method can also run with its output passed through the safety layer.

The learned methods share the attention encoder where possible. Attention+penalty is the penalty-based twin of SA-Attn: it learns from the profit minus a weight w times the shortfall and overload energy, without the safety layer. Its constrained counterpart, Attention+PPO-Lagrangian, replaces the fixed weight by Lagrange multipliers with a cost limit of zero [9, 28]. With $l_i = p_i^{\min}$, Attention+safety layer without departure bound reduces the layer to a per-step transformer projection as in [15], keeps $\beta = 0.01$ and learns the departure requirement from the penalty reward. Deep Sets+safety layer is a size-agnostic per-token network with a mean-pooled context and no attention, matched to the parameter count of SA-Attn within 10%. The penalty-trained checkpoints also run with the safety layer at deployment only, and SA-Attn and Attention+penalty also run without admission control, keeping the arrival-capped request. A multilayer perceptron (MLP) over zero-padded tokens, with and without the layer, is a fixed-size reference evaluated only at $N \leq 20$. As an in-distribution reference, SA-Attn is also trained at $N = 50$ with three seeds and evaluated at $N \in \{50, 100, 200\}$.

Six rules complete the comparison: as fast as possible, as late as possible, round robin, LLF, sLLF [18] and a price-threshold rule. Our HiGHS re-implementation of the economic MPC of EV2Gym [27] runs at every N with a 10-step horizon (eMPC-HiGHS) and a 24-h horizon (eMPC-HiGHS-24h). Both know the exact prices and arrivals inside their horizon, an information advantage, and the original Gurobi-based eMPC runs on the literature configuration as a cross-check. EV-GNN [11] runs from its public code at commit `01aa1b7a` with its own TD3 training script, reward and hyperparameters, one model per arrival profile at $N = 20$ and $\rho_{\mathrm{tr}} = 4$ kW with five seeds. Unlike our methods, it trains on its own configuration with one load and PV draw per run and with 3×10^6 steps instead of its default of 10^7. It is reported both in that configuration and zero-shot. GNN-DT [7] has public code (DT4EVs, commit `8b327d12`), but it trains offline on datasets that an optimiser generates for each scenario, and porting this pipeline

was beyond the scope of this study.

Ablations of SA-Attn set $\beta = 0$, train with a curriculum or use a smaller network. Attention+penalty is further trained with a dense per-step deficit penalty and with $w \in \{1, 100\}$ €/kWh (three seeds). The conformal variant SA-Attn-CP is trained once at $\alpha = 0.1$ and evaluated with bounds recalibrated for every α, with an uncalibrated point prediction as its ablation.

5.3 Metrics

The main metric is the penalised-profit gap of Equation 8, valid for every method, and profit, energy not served and transformer overload energy are reported separately. For the safety layer we report the share of EVs whose stay contains no infeasible step, the infeasible steps per fallback case and the measured shortfall against the deficit bound of Lemma S5. The cost of the guarantee is the difference between Attention+penalty and SA-Attn on the same episodes, in J^{pen} as the primary and in profit as the secondary measure. It is a ratio of sums over matched episodes, in € per EV served and in percent of $|J^{\text{LP,ceil}}|$. A positive value means that the penalty-based twin scored higher. The transfer gap of SA-Attn to the reference trained at $N = 50$ separates transfer loss from method loss.

5.4 Tuning, statistics and compute

Each of the seven tuned learned methods receives 12 configurations from a grid over learning rate, minibatch size and entropy coefficient. Penalty rewards add the weight $w \in \{1, 10, 100\}$ to the grid, and the Lagrangian method adds the multiplier step size. Every configuration trains on the training mixture for 0.5×10^6 steps and is evaluated on 24 validation episodes disjoint from the test replays. Selection uses the validation J^{pen} at $w = 10$ for every method, so a penalty method cannot win by lowering its own weight. The selected configuration is retrained with five seeds for 3×10^6 steps, and Table S4 lists the hyperparameters.

The training seed is the unit of analysis, the number of seeds is fixed in advance, and replay-level values are averaged within each seed first. Our pre-declared primary endpoints are the gap at $N = 200$ (E1), the cost of the guarantee per N (E2), the overload energy (E3) and the share of EVs without infeasible steps (E4). As the primary test, a paired t-test compares seed means on the same fixed replays, and a Wilcoxon signed-rank test [74] on replay-level differences of the seed-averaged policies is secondary. Holm's correction runs over each results table, and stratified-bootstrap 95% confidence intervals, the interquartile mean (IQM) and the probability of improvement are reported [75]. Results over few seeds can mislead [76], and with five seeds the tests have low power, so the confidence intervals are the primary evidence (Section S4.4).

The implementation uses PyTorch [77], a single-file PPO in the style of CleanRL [78] and the Adam optimiser [79], with the safety layer in double precision. Overall, the core experiments need about 32 GPU-hours of training and 82–132 CPU-hours for evaluation, admission tables, calibration and optima (Section S4.5). All experiments are run by the authors on their own hardware [TODO: CPU, GPU, memory, operating system, Python and PyTorch versions]. Code, configurations and analysis scripts will be released with the paper [TODO: repository link].

6 Results

[TODO: Skeleton: the authors replace every sentence frame below with their own text, fill the red placeholders from the analysis outputs and write all interpretation themselves].

6.1 Main comparison

Table 2 compares all methods at the training sizes $N \in \{10, 20\}$, pooled over profiles and ratings. At these sizes, the gap of SA-Attn is [RESULT: gap and IQM with 95% confidence interval, Table 2]. Relative to the other learned methods, the difference in penalised profit is [RESULT: differences, confidence intervals and

Table 2: Main comparison at the training station sizes $N \in \{10, 20\}$ (three arrival profiles and both tightness levels pooled; placeholder until the analysis scripts have run).

Method	J^{pen} [€]	Gap [%]	IQM gap [%]	Profit [€]	Not served [kWh]	Overload [kWh]
SA-Attn (ours)	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
Attention+penalty	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
Attention+PPO-Lagrangian	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
Deep Sets+safety layer	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]
Rules and eMPC-HiGHS	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]	[RESULT:]

RESULT FIGURE PLACEHOLDER

expected file: `fig3_zero_shot_scaling.pdf` (analysis/figures/)

Penalised profit per charger and optimality gap versus N per method, 95% confidence intervals; training sizes marked.

Figure 3: Zero-shot transfer across station sizes for N from 5 to 200.

Holm-adjusted p-values, stats/tests_T1.csv]. Among the rules and model-predictive methods, [RESULT: best rule with the safety layer and the gaps of eMPC-HiGHS, Table 2].

6.2 Zero-shot transfer across station sizes

Figure 3 plots the penalised profit per charger and the gap against N for weights trained at $N \in \{10, 20\}$. At $N = 50$, 100 and 200, the gap of SA-Attn is [RESULT: gap per N with 95% confidence interval, Table S5], and its transfer gap to the reference trained at $N = 50$ is [RESULT: Table S5]. EV-GNN reaches [RESULT: gap in its training configuration and zero-shot, Table S5], and at the stress level $\rho_{\text{tr}} = 3$ kW the gap of SA-Attn is [RESULT: Table S13].

6.3 Constraint satisfaction

Figure 4 shows energy not served, overload energy and the share of EVs without infeasible steps (E4) against N for the main methods. With the safety layer, the overload energy is [RESULT: value from Table S6; Theorem 1(a) predicts zero], and without it the methods leave [RESULT: energy not served and overload, Table S6]. E4 of SA-Attn is [RESULT: per N, Table S7], and the aggregate limits bind on [RESULT: binding rates per N, Table S7]. Infeasible steps fall into the fallback cases as [RESULT: counts per case, Table S7], and the shortfall reaches [RESULT: ratio to the deficit bound, Table S6].

6.4 Cost of the guarantee

Figure 5 plots the cost of the guarantee of Section 5.3 against N. In penalised profit, Attention+penalty minus SA-Attn is [RESULT: ΔJ^{pen} in € per EV served and in % of $|J^{\text{LP,ceil}}|$ per N, Table S8]. Measured in raw profit, the difference is [RESULT: Δ profit per N, Table S8], with the tests in [RESULT: stats/tests_T5.csv].

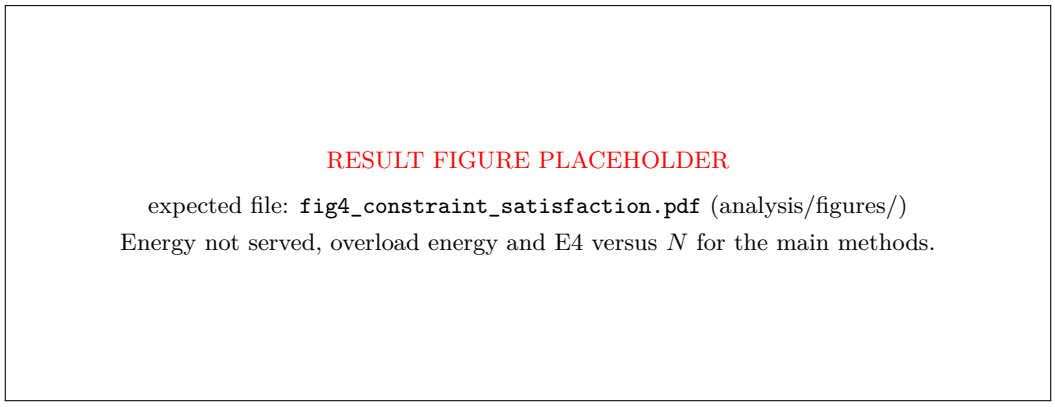


Figure 4: Constraint satisfaction across station sizes, including the share of EVs whose stay contains no infeasible step (E4).

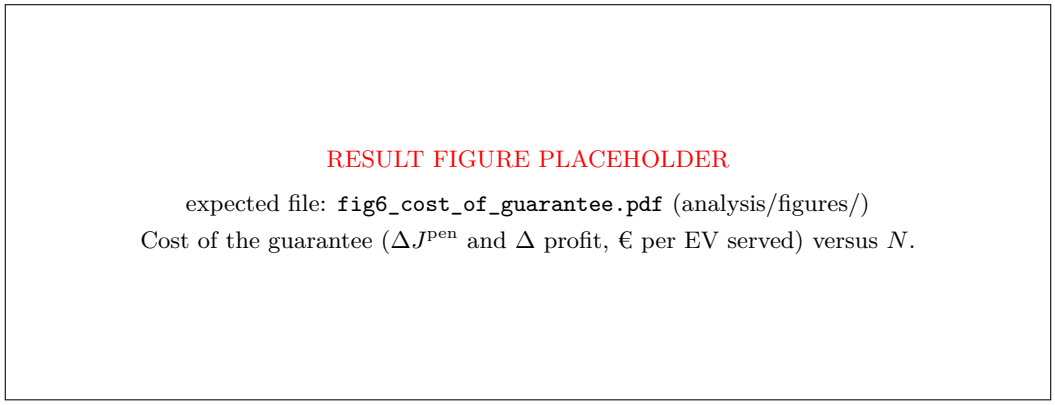


Figure 5: Cost of the guarantee across station sizes. A positive value means that Attention+penalty scored higher than SA-Attn on the same episodes.

6.5 Further results

In Table S9, the variants without departure bound, with the layer at deployment only and without admission control give [RESULT: gaps relative to SA-Attn, Table S9]. For the conformal variant, the gap and the episode-averaged coverage are [RESULT: per α, Table S10], and one control step at $N = 200$ takes [RESULT: inference time, Table S11]. The sensitivity studies give [RESULT: Table S13], and Figure S7 shows an example episode.

7 Discussion

7.1 Interpretation

[TODO: Paragraph 1, by the authors: state whether weights trained at 10 and 20 chargers transfer to 200 chargers and to the stress level. Use Figure 3, the transfer gap and fallback columns of Table S5, the binding rates of Table S7 and Table S13].

[TODO: Paragraph 2, by the authors: relate the cost of the guarantee of Figure 5 and Table S8 to the violations of Attention+penalty in Table S6. Use ΔJ^{pen} as the primary measure and stats/tests_T5.csv].

[TODO: Paragraph 3, by the authors: discuss E4 and the shortfall relative to the deficit bound in Table S6 and Table S7. Relate both to the closed-loop nature of A5].

[TODO: Paragraph 4, by the authors: compare with published EV2Gym numbers only on the literature configuration (Table S14), never across different simulators or scenarios].

7.2 Practical implications for charge point operators

This subsection is to be written by the authors after the experiments, covering the following points.

- [TODO: What a CPO gains when the transformer limit holds for every raw action, and what the conditional departure result means for the service promise.]
- [TODO: The cost of the guarantee in euros per EV served for a station of given size, from Table S8.]
- [TODO: Use of the conformal variant where drivers do not declare departure times, and acceptable coverage levels, from Table S10.]
- [TODO: Inference time per control step from Table S11 relative to the 15-min control step.]

7.3 Limitations

The departure result of Theorem 1 rests on declared departure times and requirements (A2). Our conformal extension relaxes only the departure step, its coverage is marginal, and strategic misreporting of deadlines [60] lies outside the model. Aggregate feasibility (A5) can fail in closed loop, as the two-EV example of Section 4.4 shows. On such steps the departure requirement is guaranteed only for the EVs covered by Theorem 1(b), and Lemma S5 bounds the remaining shortfall without removing it. A layer that certified joint feasibility over the future would need time-coupled constraints and would lose the single-sum projection.

The transformer result needs the one-step model A3 and the exogenous headroom of A6, and the controller does not enforce A6. Lemma S7 proves A3 for the EV2Gym settings we use, whereas a real battery management system may draw less than commanded. Under-delivery lowers the energy of an EV below the value the bounds assume, so hardware would need a margin on the thresholds or feedback from measured energy, a remedy not analysed here. A1 holds for EV2Gym's exponential taper, but a much steeper taper would break it. The model covers one station behind one transformer without voltage limits or line flows, and further coupled sum constraints would fall outside Proposition 1.

The experiments use EV2Gym's lossless accounting, and the discharge price equals the charging price. V2G arbitrage therefore carries no round-trip loss and no degradation cost, so profits from discharging may be optimistic, and the sensitivities probe this effect only at one station size. All results come from simulation with synthetic arrivals from EV2Gym's statistics and Dutch day-ahead prices, some late evening sessions do not fit into the episode horizon, and forecast noise is synthetic. Grid quantities scale linearly with N, so the per-charger statistics of the test stations match training, and only the stress level changes them.

Some comparisons are not fully like-for-like. EV-GNN trains per profile at one size with its own algorithm and reward, on one load and PV draw per run and with fewer steps than its default. Our eMPC-HiGHS re-implementation knows the exact prices and arrivals inside its horizon, the MLP is reported only at $N \leq 20$, and GNN-DT was not run. The optima use outer taper rows, exact except on a one-step transition band and a 1% margin, together with the rounding allowance. They also ignore the information limits of an online controller, so part of every gap is the price of not knowing the future.

8 Conclusion

This paper combined a size-agnostic attention policy with a feasibility-preserving safety layer for V2G charging control at a CPO station. The layer projects raw actions onto taper-aware per-EV bounds and the two-sided transformer limit in $O(N \log N)$ time, during training and deployment. Theorem 1 gives a two-layer result. Under the one-step model and the grid-headroom condition, the transformer limit holds at every step for every sequence of raw actions. Every EV whose stay contains no infeasible step meets its admitted departure requirement, and a deficit bound limits the shortfall of the others.

In the zero-shot evaluation, [RESULT: one or two sentences on the penalised-profit gap up to $N = 200$, Table S5]. Regarding constraint satisfaction, [RESULT: one sentence on overload energy, shortfall and

E4, Tables S6 and S7]. The cost of the guarantee was [RESULT: one sentence on ΔJ^{pen} and Δ profit per EV served, Table S8].

For a CPO, three limitations matter most. First, the departure result needs declared requirements and departure times, or a calibrated departure bound with marginal coverage only. Second, the transformer result assumes that each EV draws the commanded power, and real battery management systems may draw less. Third, the economics omit round-trip losses and battery degradation. Future work can add measured-energy feedback to the bounds, extend the feasible set to several transformers and linearised voltage limits, and certify joint feasibility over the future. Tests with recorded charging sessions would narrow the remaining gap between the model and practice.

Article Information Form

Authors' Contributions
Removed for double-blind review.

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Conflict of Interest
The authors declare that there is no conflict of interest regarding the publication of this paper. [TODO: Confirm this declaration].

Support/Supporting Organizations
[TODO: Name every source of financial support with grant numbers, or state that no financial support was received].

Ethical Approval
[TODO: Confirm the following wording]. This study is a simulation study that involves no human participants, no human data collected by the authors and no animals, so ethics committee approval was not required. The authors declare that they complied with research and publication ethics.

Availability
The code, the scenario configurations, the test replays and the analysis scripts will be released in a public repository. The link is withheld for double-blind review. The experiments use the public EV2Gym code at commit `55ef368c` and, for one baseline, the public EV-GNN code at commit `01aa1b7a`.

Artificial Intelligence Statement
[TODO: To be written by the authors. A factual record of the AI contributions is kept in `review/ai_contribution_record.md`].

Plagiarism Statement
This article has been scanned by iThenticate™.

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